

Example suggestions (real harness output)

Model: **gemma-4-E2B base**, Q4_K_M, Metal, greedy (temp=0), n_predict=24. Passage: work email. **shown** = what an inline UI would display (trimmed to first clause/sentence).

Aggregate over the 21-request loop: word-prefix-match mean **0.67**, histogram {'0': 12, '2': 3, '1': 5, '3': 1} (matched leading words vs the author's actual continuation — a strict lower bound).

#	text typed so far (tail)	suggestion shown	author actually wrote next
0	...ah,\n\nThanks for sending over the	details.	draft of the quarterly report yesterday. I we
1	...quarterly report yesterday. I went	through it and made some comments.	through it this morning and I think it is in
2	...ugh it this morning and I think it	looks great.	is in really good shape overall. The summary
3	...ly good shape overall. The summary	is clear and concise, and the data is well-presented.	on the first page is clear and the revenue ch
4	...on the first page is clear and the	data is presented in a way that is easy to understand.	revenue charts are much easier to read than I
5	...harts are much easier to read than	the previous version.	last time. I do have a couple of small sugge
6	...ast time.\n\nI do have a couple of	comments:	small suggestions before we send it to the wi
7	...ggestions before we send it to the	board.	wider team. First, it would be helpful to add
8	...eam. First, it would be helpful to	have a section that summarizes the key points of the report.	add a short note explaining why the marketing
9	... note explaining why the marketing	and fundraising numbers are down this quarter.	spend went up in March, since a few people as
10	...pend went up in March, since a few	people have asked about that.	people asked about that last quarter. Second,
11	...t that last quarter. Second, could	you add a note about the new marketing plan for the next quarter?	you double check the numbers in the table on
12	...ble check the numbers in the table	on page 3?	on page four? One of the totals does not seem
13	... page four? One of the totals does	not seem to be correct.	not seem to match the chart above it. Other
14	... seem to match the chart above it.	∅	Other than that, I think we are ready to go.
15	...er than that, I think we are ready	to send it to the wider team.	to go. Let me know if you want to hop on a qu
16	...ady to go. Let me know if you want	to send it to the wider team.	to hop on a quick call to walk through the ch
17	...ant to hop on a quick call to walk	through the report.	through the changes, otherwise I am happy to
18	...e changes, otherwise I am happy to	send it to the wider team.	review the next version whenever it is ready.
19	...next version whenever it is ready.	∅	Thanks again for putting this together. Bes
20	...ng this together.\n\nBest,\nDaniel	∅	

Mid-cursor (edit-in-the-middle) — the suffix-collision failure (config H)

Model: **gemma-4-E2B base**, Q4_K_M, Metal, greedy, n_predict=24, --mid-cursor. Here the cursor sits *inside* the email (the author went back to insert text), so the suffix already exists. The model still sees only the prefix — a causal LM cannot condition on the suffix — and we score the suggestion for **collision** with that already-written suffix.

Aggregate over the 19 mid-cursor points: **suffix-collision rate 0.47**, mean leading suffix words duplicated **0.68**, 3-gram verbatim re-use **0.11**. Overlap here is the *failure* signal (the opposite of the end-of-text table): the model re-proposes text the author already has just to the right of the cursor.

#	text before cursor (tail)	suggestion (model, prefix-only)	suffix already to the right	collision
1	...quarterly report yesterday. I went	through it and made some comments.	through it this morning and I think it is in re	dup 2 words

#	text before cursor (tail)	suggestion (model, prefix-only)	suffix already to the right	collision
3	...good shape overall. The summary	is clear and concise, and the data is well-presented.	on the first page is clear and the revenue char	3-gram
10	...spend went up in March, since a few	people have asked about that.	people asked about that last quarter. Second, c	dup 1 word
12	...double check the numbers in the table	on page 3?	on page four? One of the totals does not seem t	dup 2 words + contradicts (3 vs 4)
13	...page four? One of the totals does	not seem to be correct.	not seem to match the chart above it.	dup 3 words + 3-gram
17	...want to hop on a quick call to walk	through the report.	through the changes, otherwise I am happy to re	dup 2 words

Case 12 is the clearest: the model both **duplicates** the suffix ("on page...") and **contradicts** it ("page 3" vs the author's "page four"), because it has no idea the "page four" text already exists. A real fill-in-the-middle model would bridge into the suffix instead of overwriting it. See `WRITEUP.md` Q4.

Base vs instruction-tuned, identical prefixes (raw, untrimmed)

On gemma-4 the gap is subtler than on older Gemma (no hard markdown leakage), but the **register** still differs: the **base** model continues as *the writer* drafting the email; the **-it** model slips into a *helpful-reviewer* voice ("I've had a chance to review it", "a few minor suggestions"). Base is also in-distribution for raw continuation; -it is used without its chat template, so it can switch to assistant mode or stop early.

text typed so far (tail)	BASE (raw)	IT (raw)
...hanks for sending over the	details.\n\nI have a few questions for you:\n\n1. What is the nam	project brief. I've had a chance to review it, and I have a f
...y report yesterday. I went	through it and made some comments.\n\nI have a few questions ab	through it and have some initial thoughts.\n\nOverall, it's a s
...his morning and I think it	looks great. I'm not sure if you've seen the new quarterly re	's a solid start.\n\nI have a few minor suggestions, mainly arou
...shape overall. The summary	is clear and concise, and the data is well-presented.\n\nI have	is clear and the data presentation is very effective.\n\nI did
...irst page is clear and the	data is presented in a way that is easy to understand.\n\nI hav	data presentation is very effective.\n\nI did have a few minor
...e much easier to read than	the previous version.\n\nI have a few comments and questions th	the previous version.\n\nI have a few minor suggestions, mainly
....\n\nI do have a couple of	comments:\n\n1. I think the "What's Next" section is a bit too	minor suggestions, mainly around refining the language in the
...s before we send it to the	board.\n\n1. I think the "What's Next" section is a bit too lon	final approvers. Firstly, I think we could strengthen the ana
...st, it would be helpful to	have a section that summarizes the key points of the report.	add a brief section on key challenges faced during the quarte
...plaining why the marketing	and fundraising numbers are down this quarter. I think it is	spend increased this quarter. Second, and third, could we per